EHR dApp

(Electronics Health Record using ETH Blockchain)

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ABSTRACT

Blockchain are a stimulating analysis space for an extended time and therefore the edges it provides are employed by variety of varied industries. Similarly, the health care sector stands to profit vastly from the Blockchain technology thanks to security, privacy, confidentiality and decentralization. Withal, the Electronic Health Record (EHR) systems face issues relating to information security, integrity and management. We tend to gift a framework that might be used for the implementation of block chain technology in health care sector for EHR. The aim of our planned framework is first to implement Blockchain technology for EHR and second to supply secure storage of electronic records by shaping granular access rules. This framework provides the EHR system with the advantages of getting a scalable, secure and integral Blockchain-based answer.

Index Terms: Blockchain, health records, electronic health records, decentralization, and measurability.

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I. INTRODUCTION

The recent advent in technology affects all elements of human life and is dynamical the method we tend to use and understand things antecedental. simply like the changes technology has offered in numerous different sectors of life, it's additionally finding new ways that for improvement in health care sector. These edges were offered by Electronic Health However, they still face some problems relating to the protection of medical records, user possession of knowledge, information integrity etc.

This technology offers to supply a secure, transparent platform for storing medical records and different health care connected data. Before the arrival of contemporary technology, health care sector used paper based mostly system to store the medical records, i.e., victimisation written mechanism. It additionally baby-faced the problem of data-duplication and redundancy as all the establishments that patient visited had numerous copies of patient's medical records. The health care sector baby-faced a trend shift towards EHR systems that were designed to mix paper-based and electronic medical records (EMR).

The goal of EHR systems was to unravel the issues baby-faced by the paper-based health care records and to supply an economical system that may remodel the state of health care sector. The EHR systems are enforced during a variety of hospitals round the world due the advantages it provides, primarily the advance in security and its cost-effectiveness. They're thought of a significant part of health care sector because it provides a lot of practicality to the health care.

1.1 STRUCTURE OF THESIS

1.2A. INTEROPERABILITY

The EHR system additionally faces another issue that area unit as follows:

It is the method for various data systems to exchange data between them. The data ought to be exchangeable and should be usable for additional functions. A very important side of EHR systems is its Health data Exchange (HIE) or normally information sharing side. With variety of EHR systems being deployed in numerous hospitals they need a varied level of terminologies, technical and practical capabilities that makes it to possess no universally outlined normal [6].Moreover, at technical level the medical records being changed ought to be explicable, which taken piece of knowledge may well be additional used [6].

1.2B. INFORMATION ASYMMETRY

Today the best downside in health care sector outlined by the critics is data spatiality that refers to 1 party having higher access to data than the opposite party. Just in case of EHR systems or normally health care sector is affected by this downside as doctors or hospitals have access to the patient's records, so creating it central. If a patient desires to access his medical records he would have to be compelled to follow an extended and tedious method to access them.

1.2C. DATA BREACHES

Data breaches in aid sector additionally involve the requirement of a higher platform. A study [7] was in dire straits analysing {the knowledge the info the information} breaches in EHR systems and it pictured that 173 million data entries are compromised in these systems[8], explains that hospitals became a target of cyber-attacks associated an increasing trend has been witnessed by the analysers whereas conducting this study that loads of research work has been wiped out this domain furthermore, several EHR systems don't seem to be designed to satisfy the {requirements the wants} and requirements of the patients and face the problems associated

with The literature additionally suggests that use of EHRs have introduced negative consequences to science [2].These issues create it affordable to seek out a platform that may be useful in reworking aid sector to be patient-cantered, i.e.,

Blockchain. A platform that is secure, clear and it additionally provides knowledge integrity to the medical records of the patients. This paper proposes a framework that makes such a suburbanized platform that may store patient's medical records and provides access of these records to suppliers or involved people, i.e., patient.

Moreover, our planned work is assuming to solve the higher than mentioned data imbalance and knowledge breaches downside moon-faced by the EHR system. This paper is organized as follows the section II of this paper summarizes the fundamentals of Blockchain technology and its dependencies; section III narrates the connected work wiped out this domain.

1.2 AIM OF RESEARCH

1.3 A. BLOCKCHAIN TECHNOLOGIES AND DEPENDENCY

This technology was introduced by Satoshi Nakamoto for his in-style work of digital currency or crypto-currency, bitcoin. Nakamoto used blockchain technology to unravel the double outlay downside of bitcoin however before long this novel technology was getting used in several alternative applications. Blockchain may be a chain of blocks that area unit connected along and area unit unceasingly growing by storing transactions on the blocks.

This platform uses a suburbanized approach that permits the data knowledge to be distributed which each bit of distributed information or usually referred to as data have shared possession. Blockchains holds batches of transactions that area unit hashed so providing them security and that they area unit managed by peer-to-peer networks. A blockchain has sure advantages like security, anonymity, and integrity of information with no third-party intervention. These advantages create it an affordable option to store patient's medical records thereon, as a result of the innovation of technology within the aid business has created the safety variety of researchers have additionally known that exploitation blockchain technology in aid would be a possible

1.3 B . ARCHITECTURE

To understand the blockchain design allow us to use the subsequent figure one that explains the entire method of a dealings being send from a user on the blockchain network.

1. A new dealings being sent by a user on the blockchain network suggests that a replacement block is made. A block within the blockchain is employed for keeping transactions within them and these blocks area unit distributed to any or all of the connected nodes in the network. That dealings placed within a block is broadcasted to any or all of the nodes within the network. All the nodes within the network have a replica of the whole blockchain that helps them in

verification method. Once a block containing the user dealings is broadcasted to any or all of the connected nodes, they verify that the block isn't tampered by any means that. If this verification ends up in success, then the nodes add that block in their own copy of blockchain.

13 C. KEY FEATURES OF BLOCKCHAIN

1) DECENTRALIZATION

With blockchain the knowledge is distributed across the network instead of at one central purpose. This additionally makes the management of data to be distributed and handled by accord reached upon by shared input from the nodes connected on the was before network. The information that concentrates at one central purpose is currently handled by several trusty entities.

2) DATA TRANSPARENCY

Achieving knowledge transparency in any technology is to own a trust based mostly relationship between entities. The information or record at stake ought to be secured and temper proof. Any knowledge being keep on the blockchain isn't targeted at one place and isn't controlled by one node however is instead distributed across. The possession of information is currently shared and this makes it to be clear and secure from any third party intervention.

3) SECURITY AND PRIVACY

Blockchain technology uses science functions to produce security to the nodes connected on its network. It uses SHA-256 science algorithmic rule on the hashes that area unit keep on the blocks. Achieving information transparency in any technology is to own a trust primarily based relationship between entities. The info or record at stake ought to be secured and temper proof. Any information being hold on the blockchain isn't focused at one place and isn't controlled by one node however is instead distributed across.

1.3E. CHALLENGES IN BLOCKCHAIN

1) SCALABILITY AND STORAGE CAPACITY

Storage of information on the blockchain causes two main issues i.e., confidentiality and measurability. The info on the blockchain is visible to everybody that's gift on the chain this makes the info vulnerable that isn't a desired outcome for a localized platform. The info hold on the blockchain would contain patient medical record, records and workplace results

X-rays reports, magnetic resonance imaging results and lots of different reports, all of this voluminous information is to be hold on the blockchain that will extremely have an effect

2) LACK OF SOCIAL SKILLS

The manner the blockchain technology works is intelligible by only a few individuals.

This technology continues to be in its initial phases and is continually evolving. Moreover, the shift from trustworthy EHR systems to the blockchain technology would take time as hospitals, or the other health care institutes got to utterly shift their systems to blockchain.

3) LACK OF UNIVERSALLY DEFINED STANDARDS

As this technology continues to be within the initial phases and is constantly evolving thus there's no outlined customary for it. Thanks to this the implementation of this technology in health care sector would additionally take longer and energy. Because it would need certified standards from international authorities that overlook the standardization method of any technology.

2. LITERATURE REVIEW

Blockchain technology was designed by Nakamoto [13], the fundamental plan was to own a cryptographically secured and a localized currency that will be useful for monetary transactions. Eventually, this concept of blockchain was getting used in numerous different fields of life; health care sector additionally being one in all them intends to use it. Variety of analysers have dispensed the research on this space, these analysis Works concentrate on the actual fact that whether or not the concept of mistreatment blockchain for health care sector is possible or not. They additionally determine the benefits, threats, issues or challenges associated by the usage of this technology.

2.1. THEORETICAL/ANALYTICAL RESEARCH

They explained bitcoin as a peer-to-peer distributed network used for activity bitcoin transactions. They additionally outlined that proof-of-work accord algorithmic rule alongside the mining of blockchain idea. Operating systems and different necessary ideas connected with them. The authors conjointly discuss that however good contracts may be used for the new thought of parallel blockchains. They identify that reason of mistreatment good contracts in blockchain is thanks to the decentralization that's offered through the pro the basics of good contract the author explained the various layers of blockchain that mix along to stay system.

2.2. PROTOTYPE / IMPLEMENTATION RESEARCH

This study contains 2 main functions, i.e., to create, store the knowledge gathered by questionnaires and to share that data. Another profit planned by the system is that the validation that square measure supplementary on this method square measure 1st valid to be correct fixed format then square measure parsed to differentiate the personal knowledge and specific knowledge associated with form operating systems and different necessary ideas connected with them. The authors conjointly discuss that however good contracts may be used for the

new thought of parallel blockchains. They identify that reason of mistreatment good contracts in blockchain is thanks to the decentralization that's offered through the pro the basics of good contract the author explained the various layers of blockchain that mix along to stay system These layers square measure knowledge, network, consensus, discusses the design and framework followed by good contracts however it conjointly provides Associate in Nursing insight on its applications and challenges

[26] Planned a system for management of medical questionnaires and therefore the aim of this method is knowledge sha selection of information storage and sharing of medical form is to use this knowledge for additional medical and clinical analysis purposes. They stressed that it might be useful for developing EHR systems and security problems related to these

3. APPROACH

3.1. SYSTEM DESIGN

A system was conjointly a reason thanks to that author's hand-picked blockchain technology for his or her planned framework. This study contains 2 main functions, i.e., to create, store the knowledge gathered by questionnaires and to share that data.

The GUI contains all the functions that might be accessed by a the user consistent with the assigned role might use this GUI for interacting with the opposite layer of the system,

3.2. BLOCKCHAIN LAYER

The next layer on the system is that the blockchain layer; this layer contains the code or mechanism for interaction of user with the (DAPP) that is engaged on the blockchain. Is that the method by that external user will update the state of the record or data keep on the Ethereum. The blockchain network These transactions square measure treated as assets by the Ethereum. The blockchain as they're piece of knowledge that user will send to a different user or to easily store it for

3.3. SYSTEM IMPLEMENTATION

The system was enforced by mistreatment the Ethereum and its dependencies. This section explores system implementation in additional detail to induce Associate in Nursing insight on the system varied functions.

3.4. SMART CONTARCTS

As explained earlier, sensible contracts square measure a crucial a part of DAPP as they're used for performing arts basic operations. Following contracts square measure enclosed during this framework: These contracts square measure used for giving access to the users on

The DAPP and performing arts CRUD operations on the records of patient. The Patient Records smart contract is formed strictly for It performs the CRUD operations at the side of the shaping roles for access of those functions. The second contract mentioned higher than, i.e., Roles is a predefined sensible contract by the Open Zeppelin sensible contract library. This library contains many sensible contracts performing varied functionalities that might be used for making you're the explanation behind mistreatment this library was to make use of the advantages it provides, ***i.e.,*** tested and com the plus library, that could be a sub-library of the Open Zeppelin library.

4. TESTING

4.1. ALGORITHM RULE

The algorithmic rule one explains the functioning of the sensible that square measure to outline roles, add, view, update and delete records.

Solidity language employed by Ethereum and is used for creating any changes within the saved records of the patient. The validation method is finished all over again to make sure that the attested users area unit accessing this operate. The last operate of rule one is delete patient records which as evident from its name is employed for deletion of the records This operate takes the distinctive id.

5. PROPOSED PROJECT

5.1. FRAMEWORK

In this project three users are available Patients, doctors and Lab. Patients will register with the application and then create profile and give access to doctors.

- Doctors will register with the application and then login and access all those patients' records who gave permission to this doctor. Doctor will add prescription to patient profile.
- Lab person will login to application using username and password as 'Lab' and 'Lab' and after login he will upload reports of patients.
- Patients can login and download or view reports based on filtration such as All Reports or date wise reports.

In this project author describe concept to store patient medical records using block chain technology as its provide inbuilt support to secure data store in it. To store details we are using block chain ETHEREUM tool and using below code we generate solidity program

```
node_modules
public
src
.gitignore
.gitignore:Zone.Identifier
package-lock.json
package-lock.json:Zone.Identifier
package.json
package.json:Zone.Identifier
SmartContract
SmartContract.sol
SmartContract.sol:Zone.Identifier
```

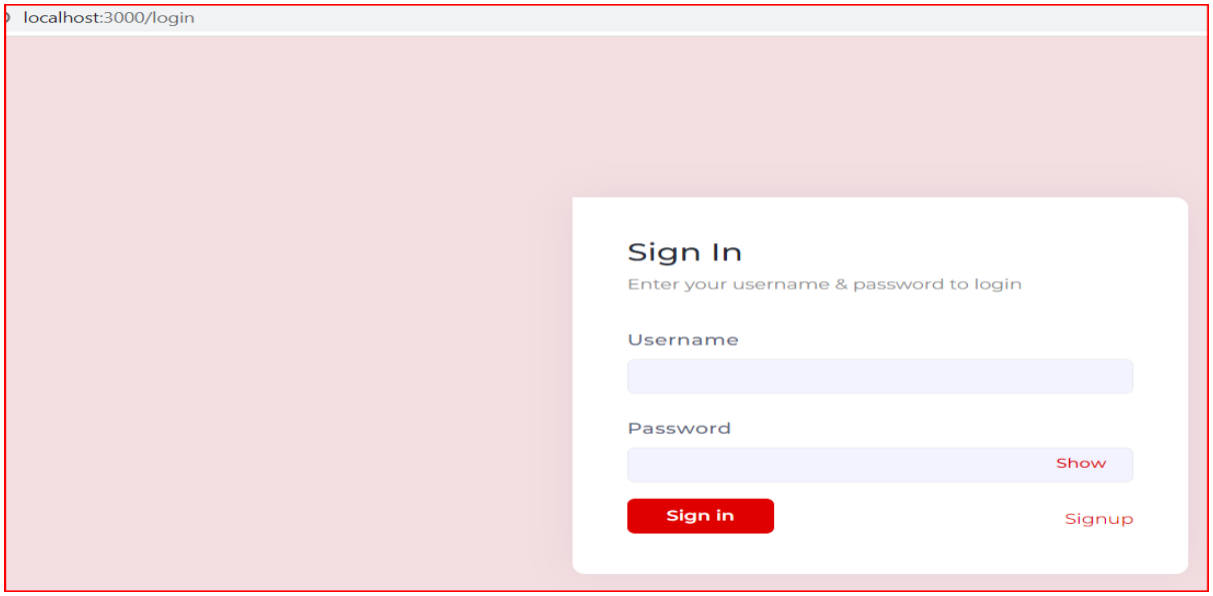
```
 *Submitted for verification at polygonscan.com
*/

//SPDX-License-Identifier:MIT
pragma solidity ^0.8.0;

contract EHR {
    struct User {
        string name;
        string username;
        bytes32 passwordHash;
        uint userType; // 1: Patient, 2: Doctor, 3: Lab Person
    }
    struct Patient{
        string username;
        string name;
        string phone;
        string symptoms;
        string description;
        string prescription;
        string reports;
    }
    struct Doctor{
        string username;
        string[] patients;
    }

    string[] public usernames;
    mapping (string => User) public users;
    mapping (string => Patient) public patients;
```

In above screen functions define to store patient, doctors and prescription details with reports.

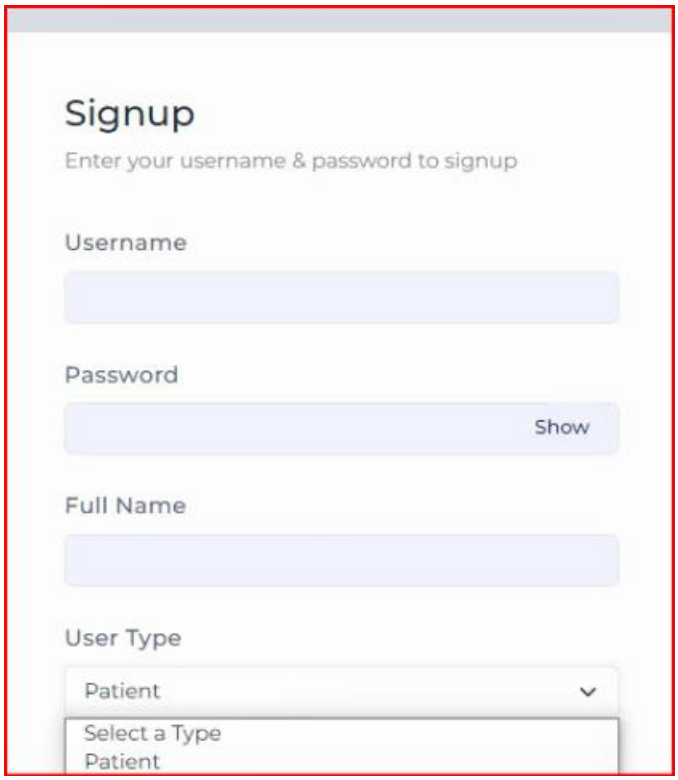
5.2 SOFTWARE INSTALLATION

Use following instructions to install NodeJS software and then start it with blockchain.

- To implement EHR DAPP on localhost, you need to do following steps:
    - Download NodeJS here, and Install it.
    - Open frontend-ehr-dapp folder in command prompt or terminal
    - Run following commands npm install OR npm install --force npm start
    - Ensure you have installed the MetaMask's in the browsers and add some eth for accessing and authentications the transactions.
    - It will automatically open the project in your default browser as below



- Download the NodeJS software and extract it and once you started it would be showing below screen in the browsers.



U need to add the smart contract and codes to the specific directory where downloaded all the codes, follow the above steps to run the NodeJS or npm start commands after running this file you must get message as smart contract deployed successfully.

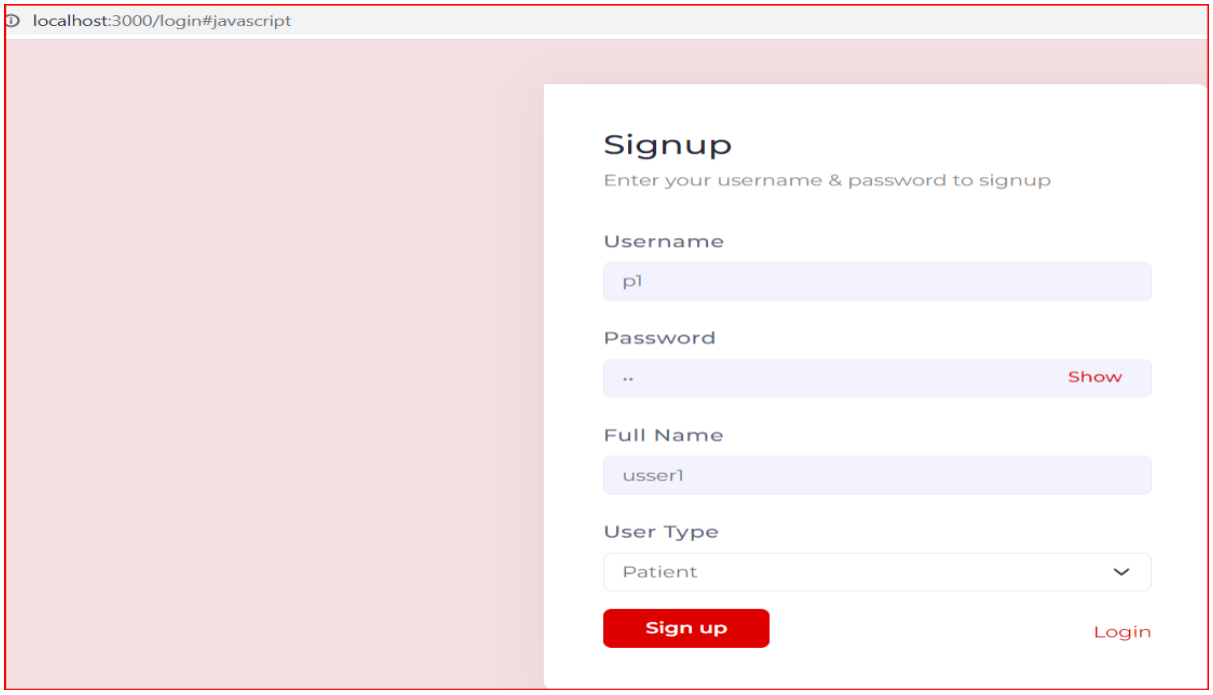
```
SmartContract > SmartContract.sol
1  /**
2   *Submitted for verification at polygonscan.com
3   */
4
5  //SPDX-License-Identifier:MIT
6  pragma solidity ^0.8.0;
7
8  contract EHR {
9      struct User {
10         string name;
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL                                   npm - Frontend

Line 604:16:  Imported JSX component Gallery_Categories must be in PascalCase or SCREAMING_SNAKE_CASE      react/jsx-pascal-case
Line 608:16:  Imported JSX component Gallery_Categories_List must be in PascalCase or SCREAMING_SNAKE_CASE  react/jsx-pascal-case
Line 612:16:  Imported JSX component Gallery_Images must be in PascalCase or SCREAMING_SNAKE_CASE          react/jsx-pascal-case
Line 616:16:  Imported JSX component Gallery_img_list must be in PascalCase or SCREAMING_SNAKE_CASE        react/jsx-pascal-case
Line 628:16:  Imported JSX component Add_Faqs must be in PascalCase or SCREAMING_SNAKE_CASE                 react/jsx-pascal-case
Line 632:16:  Imported JSX component Add_faqs_list must be in PascalCase or SCREAMING_SNAKE_CASE           react/jsx-pascal-case
Line 636:16:  Imported JSX component Career_category must be in PascalCase or SCREAMING_SNAKE_CASE         react/jsx-pascal-case
Line 640:16:  Imported JSX component Career_category_list must be in PascalCase or SCREAMING_SNAKE_CASE    react/jsx-pascal-case
Line 660:16:  Imported JSX component Contact_Page must be in PascalCase or SCREAMING_SNAKE_CASE            react/jsx-pascal-case
Line 664:16:  Imported JSX component Contact_list must be in PascalCase or SCREAMING_SNAKE_CASE            react/jsx-pascal-case
Line 668:16:  Imported JSX component Add_Category must be in PascalCase or SCREAMING_SNAKE_CASE            react/jsx-pascal-case
Line 672:16:  Imported JSX component Add_Category_list must be in PascalCase or SCREAMING_SNAKE_CASE       react/jsx-pascal-case
Line 676:16:  Imported JSX component Add_Articels must be in PascalCase or SCREAMING_SNAKE_CASE            react/jsx-pascal-case
Line 680:16:  Imported JSX component Add_Articels_list must be in PascalCase or SCREAMING_SNAKE_CASE       react/jsx-pascal-case
Line 684:16:  Imported JSX component Add_Popups must be in PascalCase or SCREAMING_SNAKE_CASE              react/jsx-pascal-case
Line 688:16:  Imported JSX component Popups_list must be in PascalCase or SCREAMING_SNAKE_CASE             react/jsx-pascal-case
Line 692:16:  Imported JSX component Add_Role must be in PascalCase or SCREAMING_SNAKE_CASE                 react/jsx-pascal-case
Line 696:16:  Imported JSX component Role_list must be in PascalCase or SCREAMING_SNAKE_CASE               react/jsx-pascal-case
Line 704:16:  Imported JSX component AddAdmin_list must be in PascalCase or SCREAMING_SNAKE_CASE           react/jsx-pascal-case
Line 709:16:  Imported JSX component All_Tickets_list must be in PascalCase or SCREAMING_SNAKE_CASE        react/jsx-pascal-case
Line 713:16:  Imported JSX component Closed_tickets_list must be in PascalCase or SCREAMING_SNAKE_CASE     react/jsx-pascal-case
Line 717:16:  Imported JSX component Open_Tickets_list must be in PascalCase or SCREAMING_SNAKE_CASE       react/jsx-pascal-case
Line 721:16:  Imported JSX component Pending_Tickets_list must be in PascalCase or SCREAMING_SNAKE_CASE    react/jsx-pascal-case
Line 730:16:  Imported JSX component Client_List must be in PascalCase or SCREAMING_SNAKE_CASE             react/jsx-pascal-case
Line 760:53:  Imported JSX component AddSocial_Links must be in PascalCase or SCREAMING_SNAKE_CASE         react/jsx-pascal-case
Line 763:49:  Imported JSX component AboutUs_page must be in PascalCase or SCREAMING_SNAKE_CASE            react/jsx-pascal-case

webpack compiled with 60 warnings
```

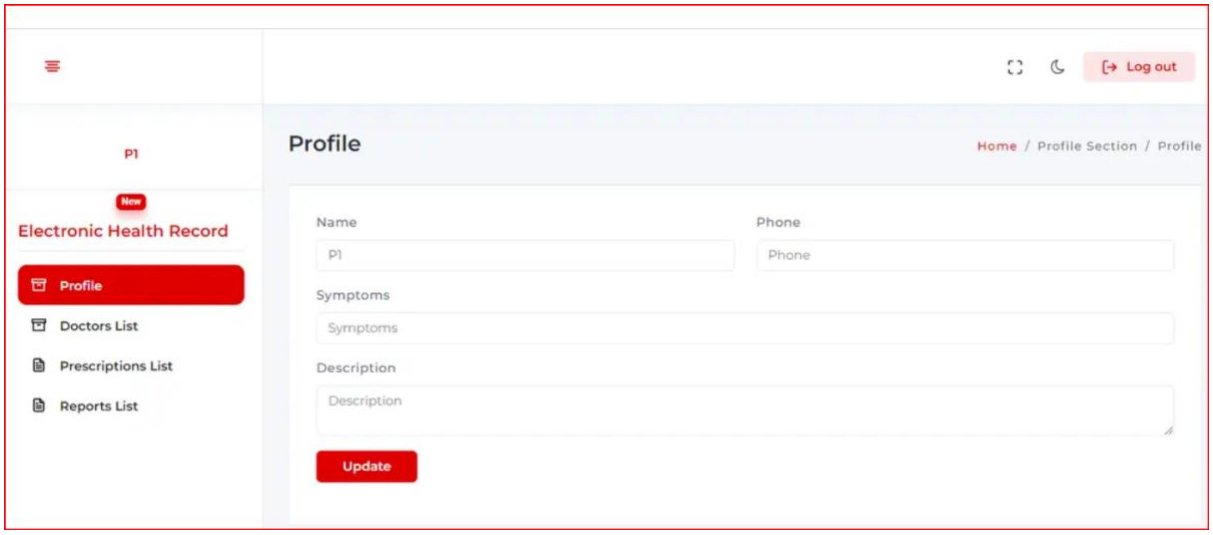
After getting that message follow below instructions to run code then open browser and enter URL as ***http://localhost:3000/login#javascript*** to get above screen

5.3 WORKING IN PROPOSED FRAMEWORK

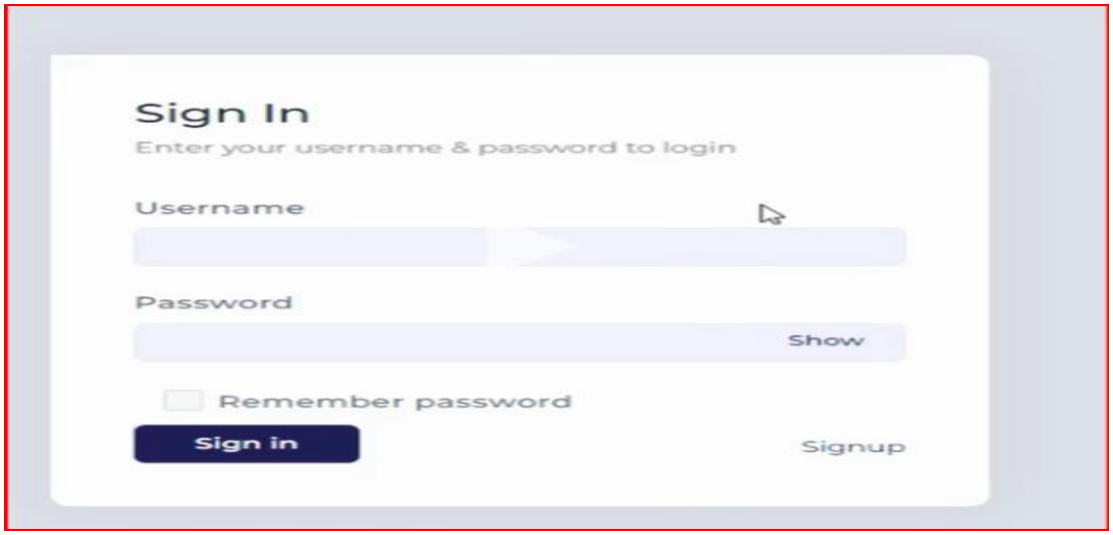
As mentioned in previous section, we've used Ethereum for implementation functions and also the expected block time Here by block time we tend to are pertaining to the number of your time taken by a replacement block For sensible contracts the number of your time it takes for a dealing to be confirmed is thirty eight seconds and this depends upon the gas value mere for the dealing. Unlike Bitcoin, Ethereum has no block size limit however instead it has a gas limit, the terms gas value and gas limit area unit The time taken for associate append operate of rule one, i.e., Add Patient Record would be around 1-2 minutes relying upon the dimensions of knowledge. For the retrieval operate like rule one operate of read Patient record would take fifty seconds.



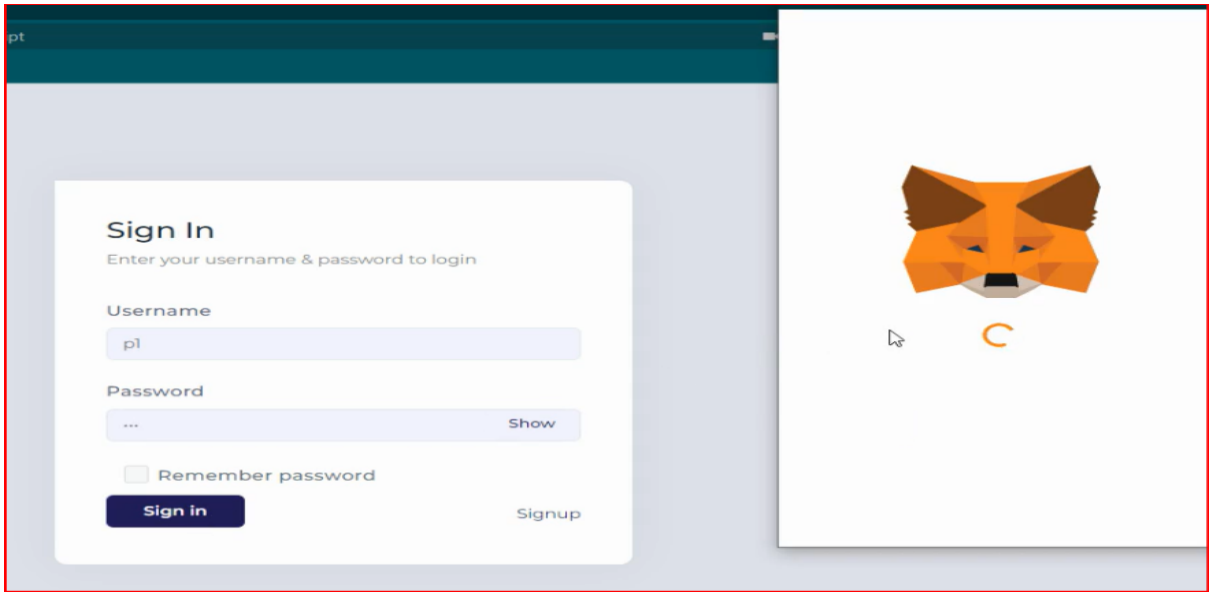
In above screen click on 'New User' and add different users



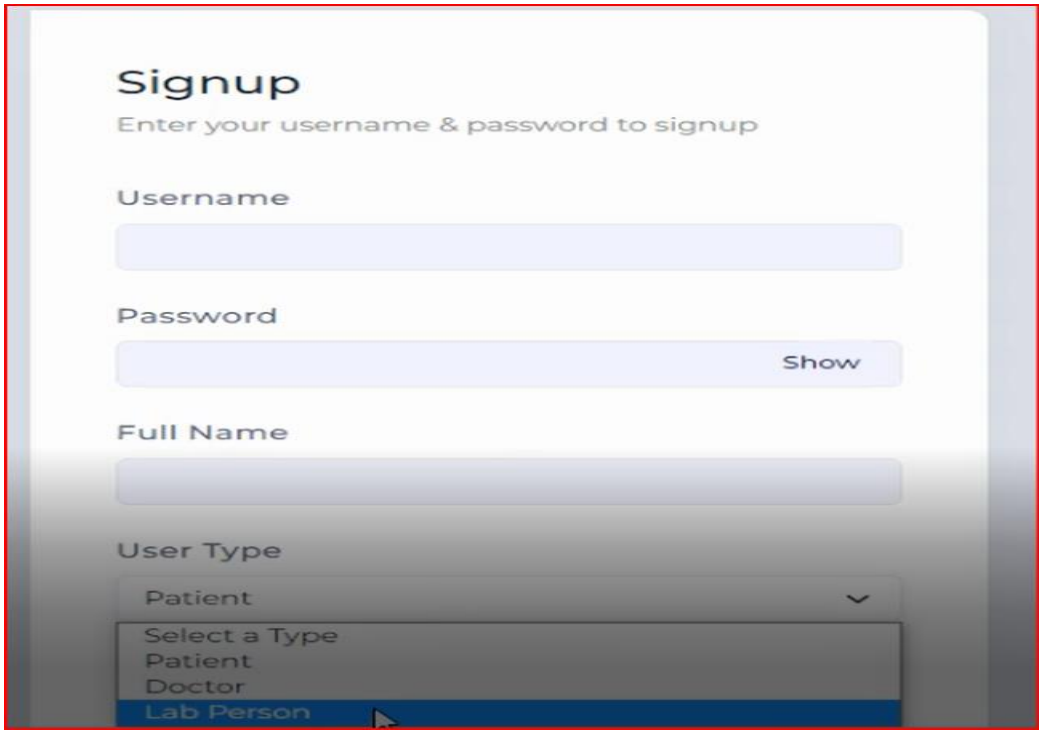
In above screen I am adding one patient details and DataOwner will be consider as patient. After adding patient details in block chain we will get below screen



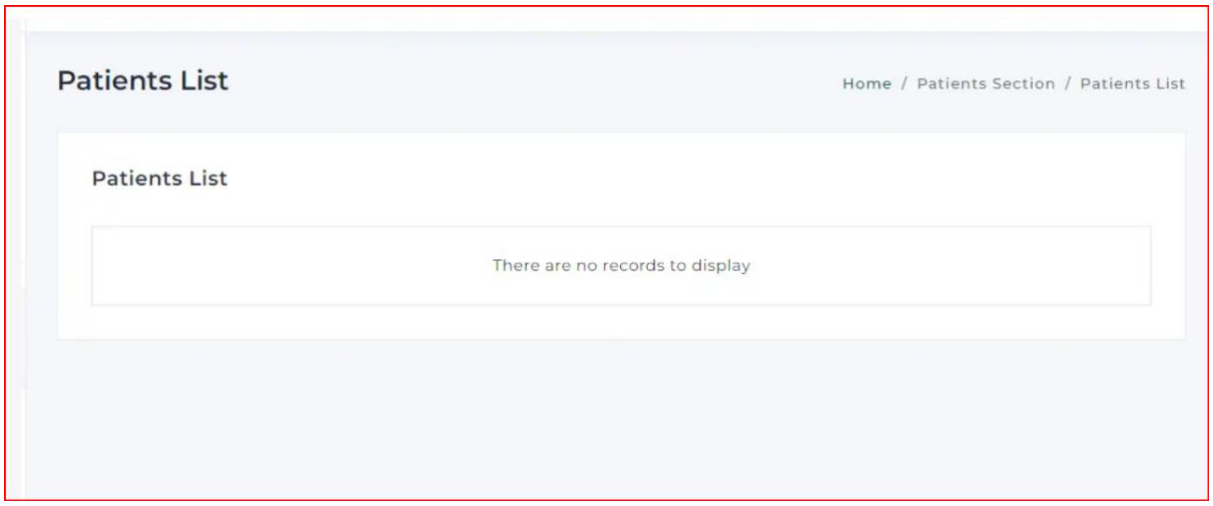
For the sing in users have to verify with the MetaMask confirmations as below –



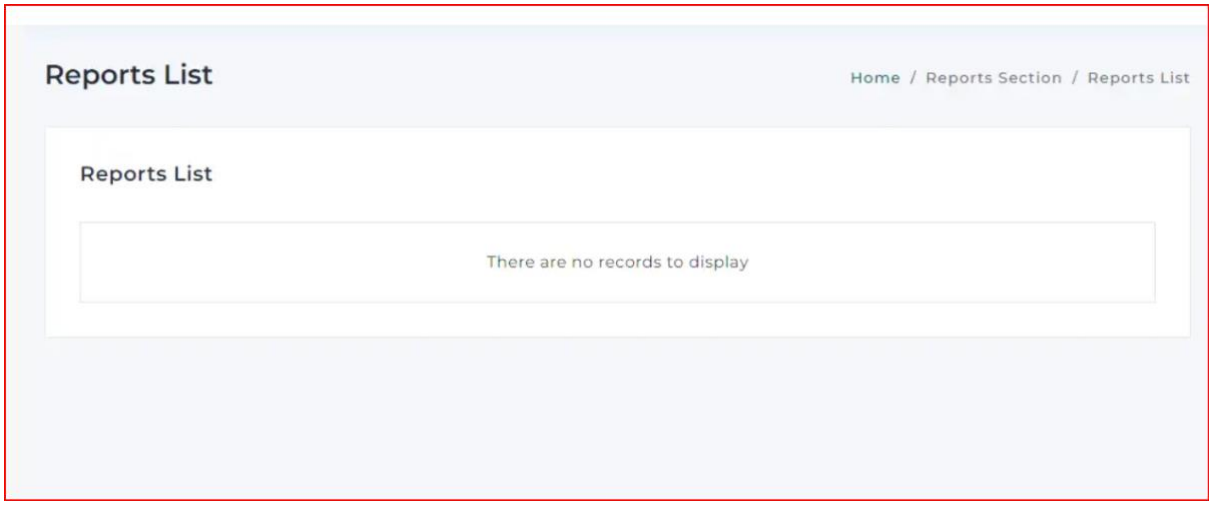
Now go back to 'New User' link again and add one doctors



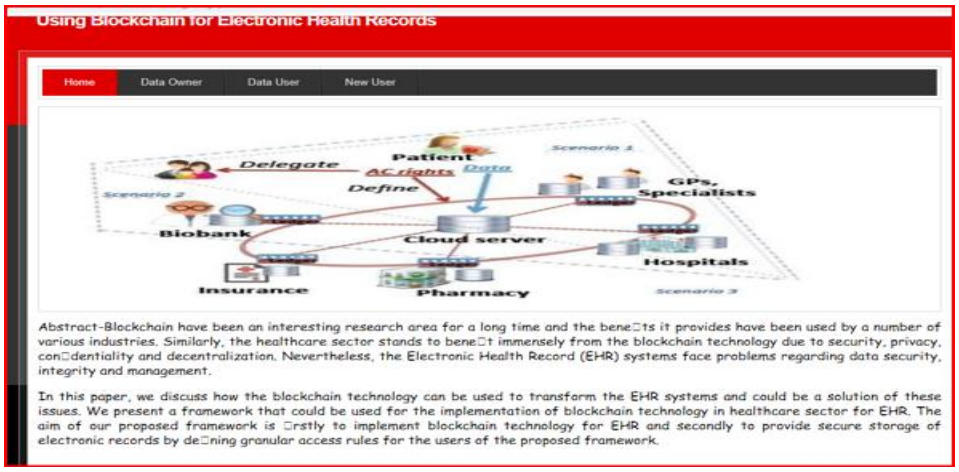
In above screen I am adding one doctor details and selected user type as 'Physician' and after adding detail will get below screen



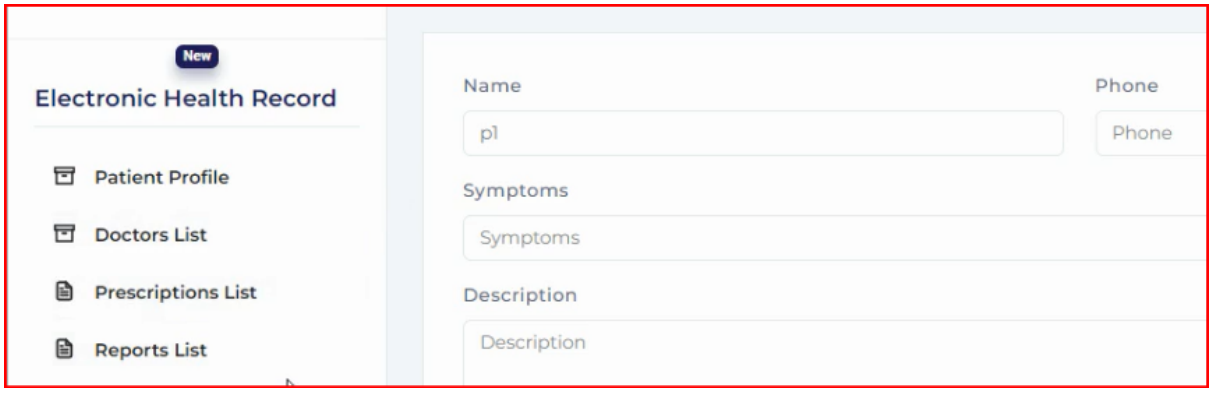
Now click on 'Data Owner' to login as patient, see below screen



In above screen I am login as patient by selecting user type as 'Data Owner' and after login will get below screen , In above screen patient will add his disease details and give access permission to doctor by selecting doctor name from drop down list. After adding disease details will get. now we got message as profile created and now click on 'View Profile' link to get details. Now, we can see all details and in last column we got message as pending which means no doctor has given any prescription. Now log out and login as doctor to give prescription to this patient

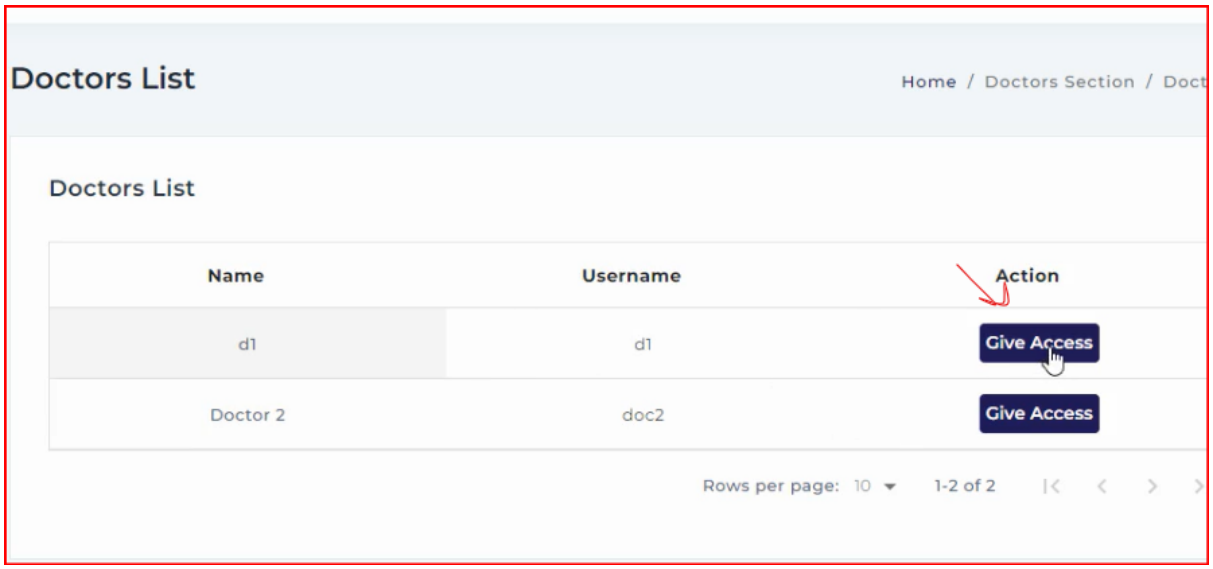


In above screen click on 'Data User' link to login as doctor In above screen I am login as doctor and after login will get below screen In above screen click on 'View Medical Profile' link to get all patient records

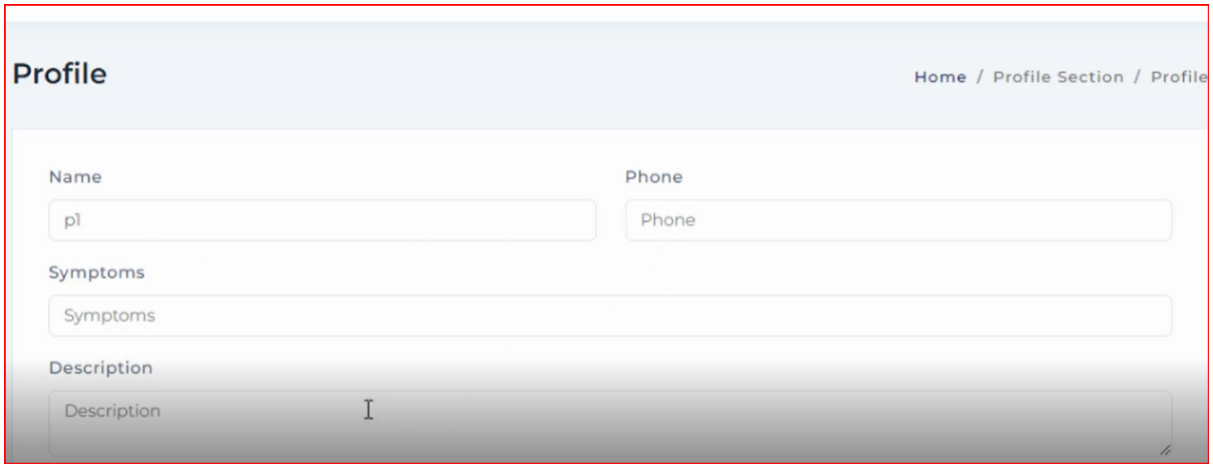


In above screen doctor will click on 'Add Prescription' link to give prescription to patient. After clicking on 'Add Prescription' link will get below screen, doctor added some prescription and patient can login and see that prescription.

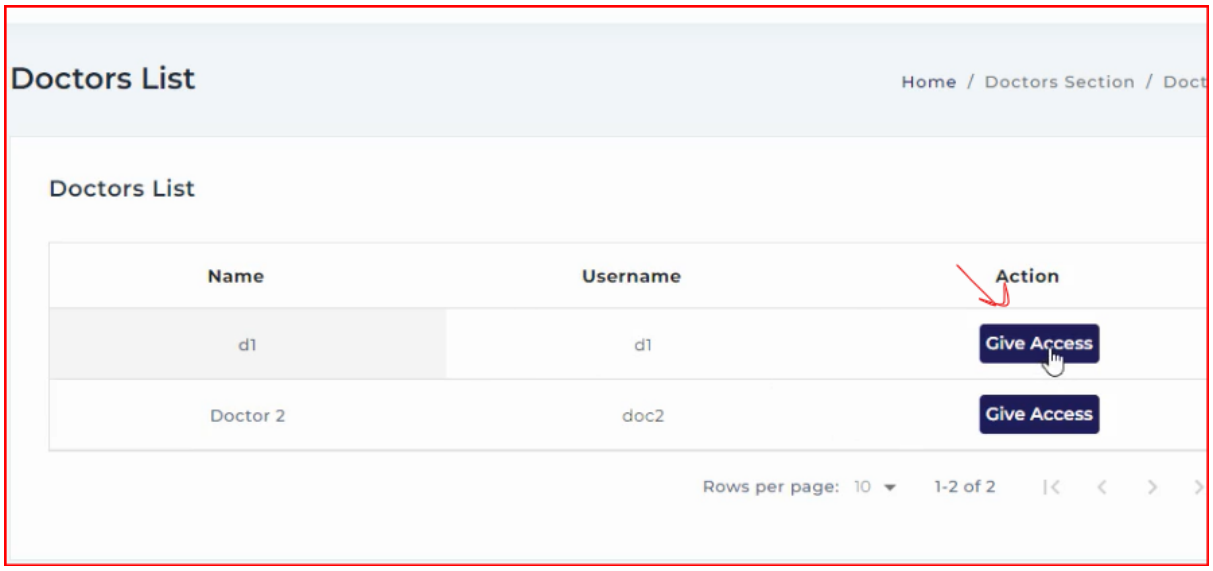
In above screen we can see status change to prescription from pending. Patient can view the prescription and go to lab for test. Now login as lab to upload report. Logout and click on 'Data User' link to get below screen now, I am login as "Lab' user and after login will get below screen



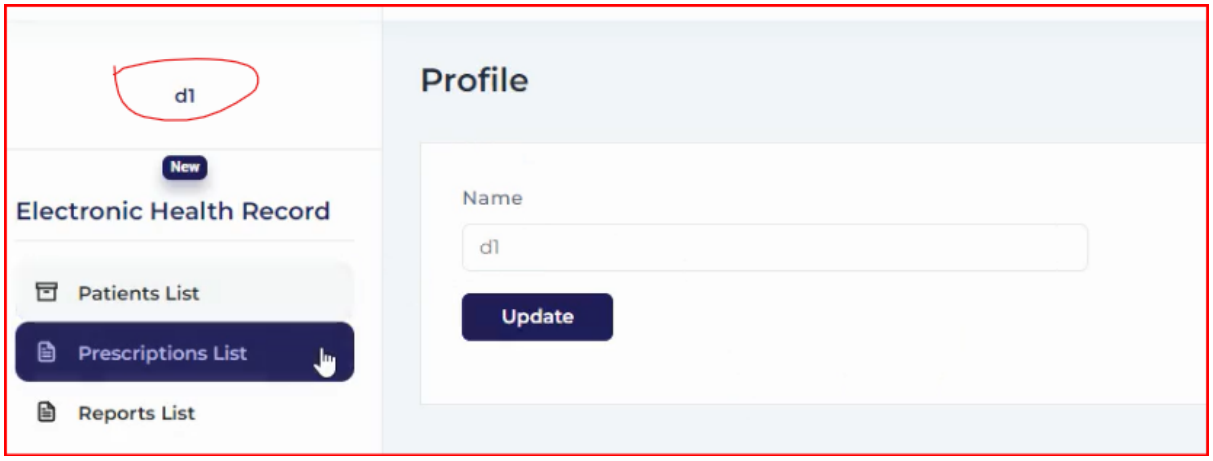
In above screen click on 'Add Reports' link to get below screen, In above screen lab person will select patient name and then upload report, In above screen I am uploading one PDF file and you can upload any type of file and after upload will get below screen



In above screen we can see report details added and now patient can login by clicking 'Data Owner' link and access reports.



In above screen patient click on 'View Reports' link to get above screen and then select either option 'All' to get all reports or select from or to date option to select all reports between two date. After selecting 'All' Option user will get below screen upon click on 'View Reports' button



In above screen patient can click on 'Click Here' link to download report , In browser status bar we can see file downloaded and similarly you can select date option and download files

5.4 FRAME WORK SCENARIO

The system chiefly has 3 entities for our projected framework they're doctor, patient and lab technicians. three users are available Patients, doctors and Lab. Patients will register with the application and then create profile and give access to doctors. After roles area unit assigned, currently once a user needs to perform success from the blockchain layer that users will read on the DAPP browser on that the full projected framework

6. ANALYSIS AND DISCUSSION OF RESULTS

6.1 PERFORMANCE

In this section we tend to evaluate the performance of the projected framework. By assessing the performance, we are able to mitigate the risks related to this novel technology that's understand-

6.2 EXPERIMENTAL SETUP

For testing performance of the projected framework we've conducted experiments by victimisation the subsequent configurations:

- Intel Core i7-6498DU central processor @ a pair of.50GHz 2.60 gig cycle per second
- And 8.00 GB of memory with Operating System Windows 10 64-bit.

6.3 DATA COLLECTION FOR PERFORMANCE EVALUATION

This section explains what quite information is employed for analysis of performance of the projected framework. This section additionally discusses the metrics that area unit accustomed justify the results of this performance analysis being conducted.

6.3. A. TRANSACTION DATA

To evaluate the performance of the projected framework following dealing information with its details area unit used. It is outlined because the time once dealing gets deployed. In Ethereum, a wise contract is deployed victimisation the transaction thus this readying time refers to it time. It is outlined because the time once the dealing is completed and Confirmed by the blockchain that during this case is Ethereum.

6.3. B. EVALUATION METRICS

The metrics used for analysis embrace the execution time, latency and turnout of the projected framework. These area units explained concisely as follows:

Execution Time is outlined as time period (in seconds)

- Throughput refers to the number of knowledge that would be transferred from one location to a different in an exceedingly unit quantity

In terms of your time it may be referred because the distinction of readying and completion time of dealing. In order to know however our projected framework would perform in real-case situation of assorted users playing

7. CONCLUSION AND FUTURE WORK

In this project we have a tendency to mentioned however blockchain technology will be helpful for tending sector and the way will it's used for care sector and technological innovation in EHR systems they still featured some problems that were addressed by this novel combination of secure record storage alongside the granular work proposes measures to make sure the system tackles the problem of information storage because it utilizes the off-chain storage and therefore the role-based access also edges the system because the medical records area unit solely out there to the For the longer term, we have a tendency to decide to implement the payment module within the existing framework. For this we want to possess sure issues as we want to make a decision what proportion a patient would acquire consultation by the doctor on this redistributed system

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